ALLOLABS

PORTFOLIO RESEARCH

Portfolio Allocation Memorandum

Comparative analysis of the Maximum Sharpe and Minimum Volatility portfolios

ANALYSIS WINDOW

2025-09-26 to 2026-06-25

PORTFOLIOS

Maximum Sharpe Model return 5.6% | Risk 11.0%

Minimum Volatility Model return 1.6% | Risk 6.2%

Prepared as an educational model memo

Expected returns are optimization inputs derived from the selected sample and research process. They are not promised or independently forecast investment returns.

CONFIDENTIAL WORKING PAPER | NOT INVESTMENT ADVICE

ALLOLABS

EXECUTIVE MEMORANDUM

Decision brief

What the two allocations are designed to do, and what matters most

---

TO Portfolio review committee

FROM AlloLabs research system

SUBJECT Allocation comparison and portfolio sensitivity review

---

This memorandum compares two portfolios built from the same eligible asset universe. The Maximum Sharpe portfolio emphasizes the strongest modeled return-to-risk trade-off. The Minimum Volatility portfolio emphasizes covariance-based risk reduction. Neither allocation should be read as a prediction; each is a conditional expression of the inputs, constraints, and historical relationships available at the run date.

PRINCIPAL OBSERVATIONS

- The Maximum Sharpe portfolio carries 11.0% model volatility and a 17.1% top-five concentration. Its largest sector is Financials at 24.9% of gross sector exposure.
- The Minimum Volatility portfolio carries 6.2% model volatility with approximately 175.0 effective holdings. Its largest sector is Financials at 16.2%.
- The largest industry allocations are Health Care (broad), Financials (broad), Consumer Discretionary (broad). Their combined behavior determines much of the portfolio's sensitivity to rates, growth, commodity prices, and earnings delivery.
- The two portfolios are alternatives, not a ranking of good versus bad. The appropriate choice depends on tolerance for concentration, model uncertainty, turnover, and adverse macro regimes.

COMPARATIVE EXHIBIT

Measure	Maximum Sharpe	Minimum Volatility
Modeled return input	5.6%	1.6%
Annualized model risk	11.0%	6.2%
Model Sharpe ratio	0.26	-0.18
Top-five absolute weight	17.1%	5.3%
Effective holdings	59.2	175.0
Active positions	108	239

PROCESS

How the portfolio is constructed

A simplified view of the analytical chain

The process converts a broad equity universe into two constrained portfolios. Each stage has a distinct purpose, and each introduces assumptions that must be monitored rather than hidden.

01

Define the opportunity set

Select eligible securities, currencies, listings, and the training window. Missing or low-quality data are removed before optimization.

02

Estimate market relationships

Calculate historical returns and covariance. These estimates describe the sample; they do not guarantee that future relationships will remain stable.

03

Form conditional return inputs

Blend equilibrium information with optional AI-assisted company views and confidence levels through the Black-Litterman framework.

04

Apply portfolio controls

Impose long-only or long/short rules, position caps, sector limits, and optional regularization before solving with SLSQP.

05

Compare and monitor

Review concentration, industry exposures, sensitivities, and out-of-sample evidence. Re-run when inputs or the investment thesis materially change.

INTERPRETIVE DISCIPLINE

The report separates measured exposure from inferred behavior. Factor sensitivities and hedge relationships are qualitative diagnostics based on sector and industry structure; they are not substitute estimates for a full scenario model, option exposure analysis, or forward covariance forecast.

Allocation comparison

Concentration, position structure, and sector composition

EXHIBIT 1 | LARGEST ABSOLUTE POSITIONS

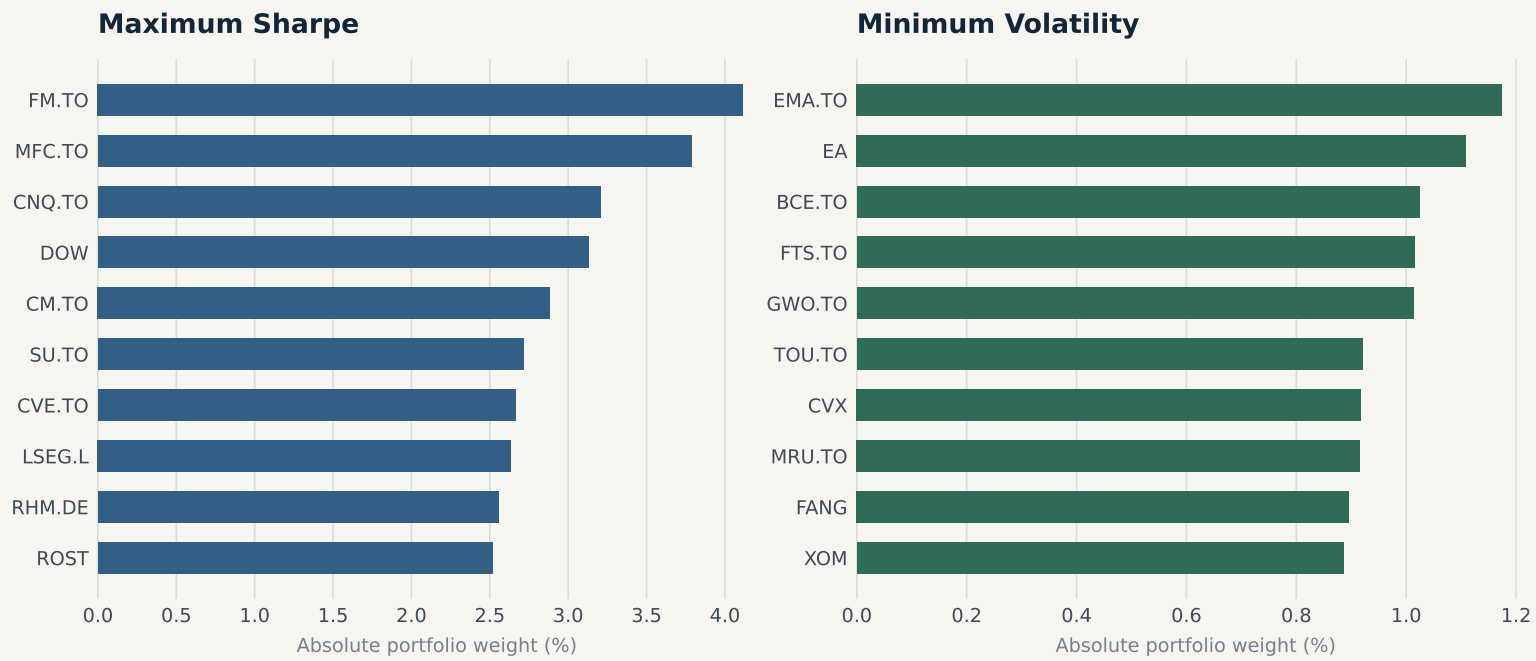
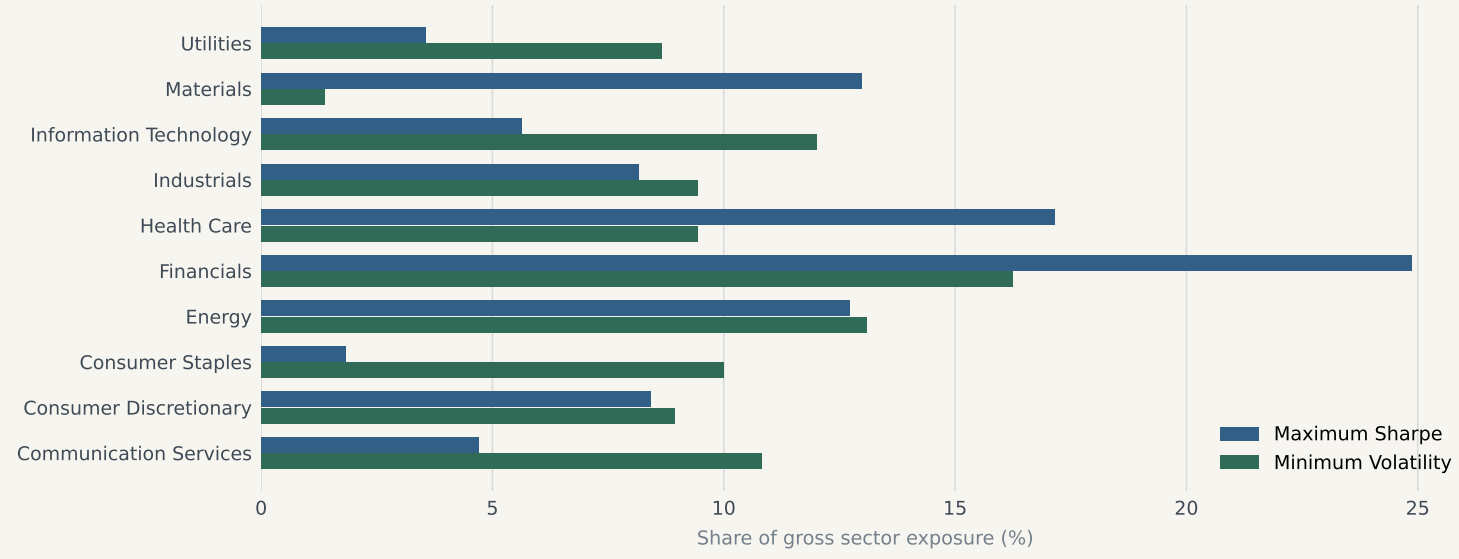


EXHIBIT 2 | SECTOR EXPOSURE



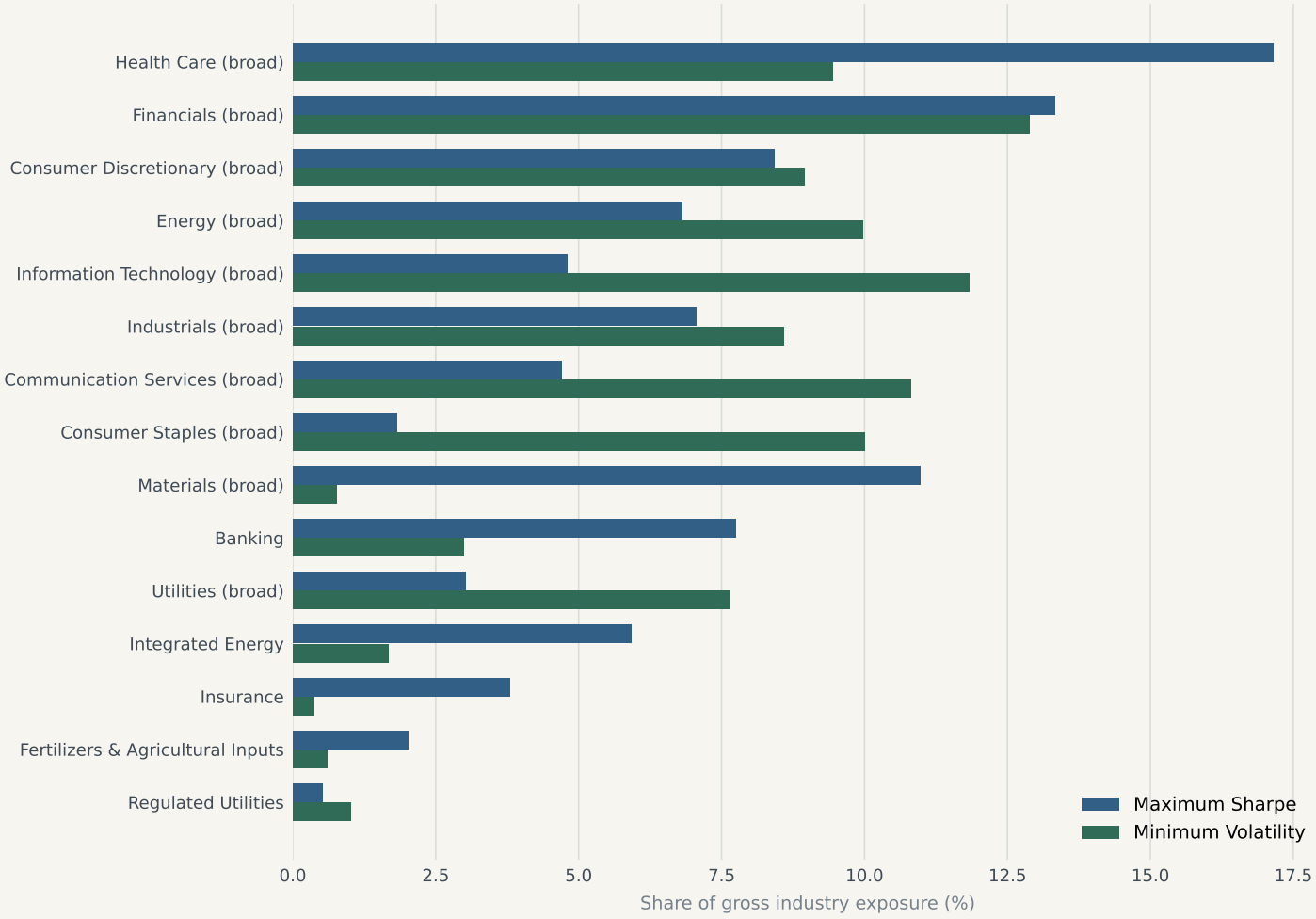
Maximum Sharpe: 17.1% in the five largest positions; 59.2 effective holdings.

Minimum Volatility: 5.3% in the five largest positions; 175.0 effective holdings.

INDUSTRY ANALYSIS

Industry allocation

Where the portfolios concentrate their underlying business exposure



INTERPRETATION

Industry exposure is calculated from absolute position weights, then normalized within each portfolio. This view reveals the business drivers beneath broad sector labels. A concentrated industry can dominate portfolio behavior even when the sector total appears diversified.

READING THE EXHIBIT

- Long and short positions both contribute to gross exposure; direction still matters when interpreting a specific security.
- Industry labels come from stored company research and fall back to the broader sector when detailed classification is unavailable.
- The next section describes the three largest combined industry exposures and their usual sensitivities.

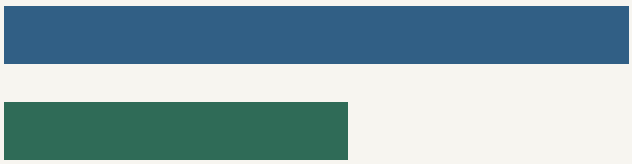
INDUSTRY ANALYSIS

1. Health care and life sciences

Health Care (broad) | Health Care | Exposure and operating sensitivity

Industry exposure profile

STRUCTURAL EXPOSURE



Maximum Sharpe 17.1%

Minimum Volatility 9.4%

Average gross signal 13.3%

GENERAL BEHAVIOR

Demand is generally less cyclical, while product pipelines, patents, reimbursement, regulation, and clinical outcomes drive company dispersion.

PRIMARY SENSITIVITIES

Economic weakness is usually less damaging than for discretionary sectors. Patent cliffs, trial failures, pricing reform, and reimbursement changes remain material idiosyncratic risks.

ROLE IN THIS PORTFOLIO

A defensive growth sleeve that can offset consumer and industrial cyclicality without relying exclusively on bond-like income.

INDICATORS TO WATCH

Pipeline milestones, patent exposure, reimbursement, procedure volumes, organic growth, and research productivity.

COMMITTEE INTERPRETATION

Treat this profile as a monitoring framework, not a forecast. Company-specific balance sheets, geographic exposure, valuation, catalysts, and position direction can materially alter the portfolio outcome.

2. Banks and capital markets

Financials (broad) | Financials | Exposure and operating sensitivity

Industry exposure profile

STRUCTURAL EXPOSURE



Maximum Sharpe 13.3%

Minimum Volatility 12.9%

Average gross signal 13.1%

GENERAL BEHAVIOR

Earnings typically respond to net-interest margins, credit quality, loan growth, trading activity, and capital requirements.

PRIMARY SENSITIVITIES

Moderately higher rates can support margins, but abrupt tightening may weaken borrowers and increase provisions. Recession risk and funding stress are the principal downside channels.

ROLE IN THIS PORTFOLIO

A cyclical income sleeve that may diversify long-duration growth assets, although it is not defensive during a credit shock.

INDICATORS TO WATCH

Yield-curve shape, deposit costs, delinquencies, loan-loss provisions, and regulatory capital.

COMMITTEE INTERPRETATION

Treat this profile as a monitoring framework, not a forecast. Company-specific balance sheets, geographic exposure, valuation, catalysts, and position direction can materially alter the portfolio outcome.

3. Consumer discretionary

Consumer Discretionary (broad) | Consumer Discretionary | Exposure and operating sensitivity

Industry exposure profile

STRUCTURAL EXPOSURE



Maximum Sharpe 8.4%

Minimum Volatility 8.9%

Average gross signal 8.7%

GENERAL BEHAVIOR

Results depend on household income, employment, credit availability, traffic, ticket size, and inventory execution.

PRIMARY SENSITIVITIES

Lower rates and improving confidence can help demand. Inflation, unemployment, weak real wages, or excess inventory can quickly pressure margins.

ROLE IN THIS PORTFOLIO

A cyclical growth sleeve that benefits from healthy consumers but needs defensive balance elsewhere in the portfolio.

INDICATORS TO WATCH

Comparable sales, traffic, average ticket, inventory, promotional activity, consumer credit, and margin guidance.

COMMITTEE INTERPRETATION

Treat this profile as a monitoring framework, not a forecast. Company-specific balance sheets, geographic exposure, valuation, catalysts, and position direction can materially alter the portfolio outcome.

PORTFOLIO INTERACTION

Structural offsets within the portfolios

Which exposures may hedge one another, by how much, and why

A hedge is not assumed merely because two industries are different. The pairings below are structural offsets: they respond to distinct economic channels and may reduce dependence on one scenario. They can still decline together during broad market stress.

1. Credit cycle versus defensive demand

MODERATELY BALANCED

Financials 20.6% **Health Care 13.3%**

Health-care demand is less tied to lending and employment, providing a different earnings engine when credit conditions deteriorate.

2. Capital cycle versus defensive growth

STRONGLY BALANCED

Industrials 8.8% **Health Care 13.3%**

Industrials depend on orders and utilization; health care relies more on clinical demand and product cycles, reducing dependence on one macro pathway.

3. Commodity producer versus commodity consumer

STRONGLY BALANCED

Energy 12.9% **Consumer Discretionary 8.7%**

Higher energy prices can support producers while pressuring household spending and transport costs; falling prices can reverse the relationship.

4. Growth duration versus regulated cash flow

STRONGLY BALANCED

Information Technology 8.8% **Utilities 6.1%**

Utilities may cushion a growth slowdown, while technology provides upside when innovation spending and risk appetite improve. Both can still be hurt by sharply higher rates.

Sizing note: percentages are the average gross sector shares across the two portfolios. Balance describes relative sleeve size, not a statistical hedge ratio.

RISK ASSESSMENT

Macro sensitivity map

How rates, inflation, growth, commodities, and risk appetite may influence the current sector mix

Indicator / shock	Max score	Max interpretation	Min score	Min interpretation
Policy rates rise	-0.19	Moderate sensitivity	-0.42	Moderate sensitivity
Inflation reaccelerates	+0.35	Mild support	+0.07	Mixed / limited
Economic growth slows	-0.57	High sensitivity	-0.36	Moderate sensitivity
Commodity prices rise	+0.35	Mild support	+0.09	Mixed / limited
Risk appetite contracts	-0.32	Moderate sensitivity	-0.30	Moderate sensitivity

LARGEST MODELED LEVERAGE

Maximum Sharpe: Economic growth slows is the largest adverse structural sensitivity in this set (score -0.57). The reading follows from the current sector mix and does not estimate the size of a market loss.

Minimum Volatility: Policy rates rise is the largest adverse structural sensitivity in this set (score -0.42). Lower covariance-based volatility does not remove event or fundamental risk.

HOW TO USE THE SCORES

- Scores range approximately from -2 to +2 and aggregate hardcoded sector sensitivities using current gross sector weights.
- Negative values indicate potential vulnerability; positive values indicate potential support. Values near zero indicate mixed or limited portfolio-level exposure.
- These diagnostics help prioritize monitoring and scenario analysis. They are not return forecasts, stress-test losses, or causal estimates.

RISK ASSESSMENT

Micro sensitivity map

How earnings, margins, capital spending, credit, and regulation may transmit through holdings

Indicator / shock	Max score	Max interpretation	Min score	Min interpretation
Portfolio earnings miss	-0.86	High sensitivity	-0.89	High sensitivity
Operating margins compress	-0.45	Moderate sensitivity	-0.56	High sensitivity
Capital spending accelerates	+0.35	Mild support	+0.36	Mild support
Credit losses increase	-0.59	High sensitivity	-0.43	Moderate sensitivity
Regulatory pressure rises	-0.61	High sensitivity	-0.58	High sensitivity

LARGEST MODELED LEVERAGE

Maximum Sharpe: Portfolio earnings miss is the largest adverse structural sensitivity in this set (score -0.86). The reading follows from the current sector mix and does not estimate the size of a market loss.

Minimum Volatility: Portfolio earnings miss is the largest adverse structural sensitivity in this set (score -0.89). Lower covariance-based volatility does not remove event or fundamental risk.

HOW TO USE THE SCORES

- Scores range approximately from -2 to +2 and aggregate hardcoded sector sensitivities using current gross sector weights.
- Negative values indicate potential vulnerability; positive values indicate potential support. Values near zero indicate mixed or limited portfolio-level exposure.
- These diagnostics help prioritize monitoring and scenario analysis. They are not return forecasts, stress-test losses, or causal estimates.

SECURITY ANALYSIS

Leading security briefs 1

Largest positions across both portfolios, with stored AI research summaries

01 FM.TO

MAX 4.1% MIN 0.0%

Materials | Not classified

No AI research summary was stored for this security in the completed run. Interpret the allocation using the measured weight, sector exposure, and underlying source data.

Posterior input: 14.7% | Research confidence: N/A

02 MFC.TO

MAX 3.8% MIN 0.4%

Financials | Insurance

Manulife’s setup is constructive, with reasonable valuation, strong reported EPS rebound, and substantial free cash flow supporting earnings normalization. Upside would come from continued profitability improvement and product traction through John Hancock. The main risk is regulatory or product scrutiny after the withdrawal of a leveraged Hong Kong offering, while mid-tier ROE tempers enthusiasm; the expected return is 0.10 with moderate confidence.

Posterior input: 7.8% | Research confidence: 73%

03 CNQ.TO

MAX 3.2% MIN 0.8%

Energy | Integrated Energy

CNQ offers a positive but cyclical return profile, with low valuation, strong ROE, and free cash flow offset by a sharp reported EPS decline and mixed recent performance. Upside would come from valuation support and resilient cash generation if commodity prices hold. The main risk is that lower earnings momentum persists or oil prices weaken, supporting a 0.08 expected return with moderate confidence.

Posterior input: 5.9% | Research confidence: 72%

AI summaries are research inputs, not verified company guidance. Review source fundamentals and current disclosures before relying on any statement.

Leading security briefs 2

Largest positions across both portfolios, with stored AI research summaries

04 DOW

MAX 3.1% MIN 0.0%

Materials | Not classified

No AI research summary was stored for this security in the completed run. Interpret the allocation using the measured weight, sector exposure, and underlying source data.

Posterior input: 8.6% | Research confidence: N/A

05 CM.TO

MAX 2.9% MIN 0.4%

Financials | Banking

CM screens as one of the more attractive bank setups, combining relatively undemanding valuation with strong ROE, reported EPS growth, and a sector tailwind from lower capital- buffer requirements. Upside would come from continued earnings growth and multiple support if credit remains stable. The primary risk is higher cyclical sensitivity in a weaker credit environment, keeping the expected return constructive at 0.09 with moderate confidence.

Posterior input: 6.6% | Research confidence: 73%

06 SU.TO

MAX 2.7% MIN 0.9%

Energy | Integrated Energy

Suncor’s 12-month setup remains constructive, supported by low relative valuation metrics, low leverage, positive free cash flow, and evidence of robust funds from operations and shareholder returns. Upside would come from continued cash generation and disciplined capital returns. The key risk is commodity-price sensitivity after a strong rally and more neutral analyst tone, limiting the expected return to 0.09 with moderate confidence.

Posterior input: 5.4% | Research confidence: 74%

AI summaries are research inputs, not verified company guidance. Review source fundamentals and current disclosures before relying on any statement.

SECURITY ANALYSIS

Leading security briefs 3

Largest positions across both portfolios, with stored AI research summaries



07 CVE.TO

MAX 2.7% **MIN 0.6%**

Energy | Not classified

No AI research summary was stored for this security in the completed run. Interpret the allocation using the measured weight, sector exposure, and underlying source data.

Posterior input: 6.1% | **Research confidence: N/A**



08 LSEG.L

MAX 2.6% **MIN 0.2%**

Financials | Not classified

No AI research summary was stored for this security in the completed run. Interpret the allocation using the measured weight, sector exposure, and underlying source data.

Posterior input: 5.5% | **Research confidence: N/A**



09 EMA.TO

MAX 0.0% **MIN 1.2%**

Utilities | Not classified

No AI research summary was stored for this security in the completed run. Interpret the allocation using the measured weight, sector exposure, and underlying source data.

Posterior input: 0.5% | **Research confidence: N/A**

AI summaries are research inputs, not verified company guidance. Review source fundamentals and current disclosures before relying on any statement.

CONCLUSION

Portfolio interpretation

A concise synthesis of the allocation, its trade-offs, and the next monitoring priorities

The Maximum Sharpe and Minimum Volatility portfolios express different objectives from the same analytical foundation. The first accepts more concentration and model dependence in pursuit of a stronger return-to-risk input. The second gives greater weight to covariance-based stability. The report does not establish that either portfolio will outperform; it clarifies what each portfolio is asking the market to deliver.

KEY CONCLUSIONS

- Maximum Sharpe combines 11.0% model risk with 17.1% in its five largest absolute positions. Its outcome is more dependent on concentrated security and industry theses.
- Minimum Volatility reduces modeled risk to 6.2% and distributes exposure across approximately 175.0 effective holdings, but remains vulnerable to regime shifts in covariance and defensive-sector valuation.
- Industry pairings provide partial structural offsets, not guaranteed hedges. Broad liquidity shocks, valuation compression, and correlated earnings revisions can overwhelm normal diversification relationships.
- The highest-priority monitoring items are the most adverse macro and micro sensitivity scores, top-holding earnings delivery, sector cap utilization, and changes in the assumptions embedded in AI views.

PRACTICAL REVIEW TRIGGERS

- A material change in rates, inflation, credit conditions, or commodity prices relative to the assumptions used at the run date.
- A top holding issues guidance or results that invalidate the stored research rationale or materially change free-cash-flow expectations.
- Concentration rises because of market movement, or a sector/position approaches its hard limit.
- Out-of-sample behavior diverges from the covariance structure used to construct the portfolio.
- The investment committee's risk tolerance, horizon, liquidity needs, or permitted universe changes.

BOTTOM LINE **Use the portfolios as disciplined starting points for review, not as automatic trading instructions.**

APPENDIX

Methodology and limitations

What the report measures, what it infers, and what it does not claim

Return inputs

Portfolio returns shown in this report are optimizer inputs derived from the selected training sample and optional Black-Litterman research views. They are not realized-return forecasts, guarantees, or independently calibrated capital-market assumptions.

Risk model

Annualized volatility and diversification statistics depend on the covariance estimate used in the completed run. Correlations can rise during stress, and historical covariance may not represent the next regime.

Industry and factor interpretation

Industry profiles, hedge relationships, and sensitivity scores are deterministic educational diagnostics based on current gross exposure and hardcoded behavioral mappings. They do not estimate loss magnitude or statistical hedge ratios.

AI research

Stored AI opinions summarize supplied fundamentals and news context. They can be incomplete, stale, or incorrect. Confidence scores represent model-assigned research confidence, not probability of positive returns.

Portfolio implementation

Weights omit taxes, commissions, bid-ask spreads, market impact, account constraints, and investor-specific suitability unless separately represented in the optimization configuration.

Use of this memorandum

The document is intended for research and education. It should be paired with current issuer disclosures, independent data verification, scenario analysis, and professional judgment.

RUN DIAGNOSTICS

Maximum Sharpe: gross 100.0%, net 100.0%, short sleeve 0.0%.

Minimum Volatility: gross 100.0%, net 100.0%, short sleeve 0.0%.

APPENDIX

Maximum Sharpe allocation schedule 1/3

All positions with absolute portfolio weight of at least 0.005%

Rank	Ticker	Sector	Industry	Weight	BL input
1	FM.TO	Materials	Not classified	4.115%	14.7%
2	MFC.TO	Financials	Insurance	3.787%	7.8%
3	CNQ.TO	Energy	Integrated Energy	3.205%	5.9%
4	DOW	Materials	Not classified	3.129%	8.6%
5	CM.TO	Financials	Banking	2.884%	6.6%
6	SU.TO	Energy	Integrated Energy	2.717%	5.4%
7	CVE.TO	Energy	Not classified	2.664%	6.1%
8	LSEG.L	Financials	Not classified	2.635%	5.5%
9	RHM.DE	Industrials	Not classified	2.556%	7.9%
10	ROST	Consumer Discretionary	Not classified	2.520%	6.6%
11	BNS.TO	Financials	Banking	2.394%	5.4%
12	FFH.TO	Financials	Not classified	2.037%	4.6%
13	NTR.TO	Materials	Fertilizers & Agricultural Inputs	2.009%	5.6%
14	ENGI.PA	Utilities	Not classified	1.892%	5.2%
15	BMY	Health Care	Not classified	1.855%	5.6%
16	TECK-B.TO	Materials	Not classified	1.651%	10.2%
17	ABT	Health Care	Not classified	1.634%	4.4%
18	BNP.PA	Financials	Not classified	1.605%	5.1%
19	ALNY	Health Care	Not classified	1.592%	5.6%
20	NOVN.SW	Health Care	Not classified	1.534%	4.6%
21	QSR.TO	Consumer Discretionary	Not classified	1.311%	3.5%
22	CHTR	Communication Services	Not classified	1.302%	5.8%
23	POW.TO	Financials	Not classified	1.297%	3.6%
24	RY.TO	Financials	Banking	1.279%	4.1%
25	GSK.L	Health Care	Not classified	1.217%	4.2%
26	BP.L	Energy	Not classified	1.183%	3.1%
27	MCHP	Information Technology	Not classified	1.144%	8.1%
28	RIO.L	Materials	Not classified	1.119%	5.8%
29	CNR.TO	Industrials	Railroads	1.111%	4.3%
30	DHR	Health Care	Not classified	1.080%	4.4%
31	KLAC	Information Technology	Not classified	1.069%	8.8%
32	GIB-A.TO	Information Technology	Not classified	1.012%	2.8%
33	PHIA.AS	Health Care	Not classified	0.998%	4.4%
34	GWO.TO	Financials	Not classified	0.998%	2.3%
35	DE	Industrials	Not classified	0.984%	5.1%
36	UBSG.SW	Financials	Not classified	0.962%	3.9%
37	BIIB	Health Care	Not classified	0.936%	3.9%
38	SONY	Communication Services	Not classified	0.926%	5.2%

APPENDIX

Maximum Sharpe allocation schedule 2/3

All positions with absolute portfolio weight of at least 0.005%

Rank	Ticker	Sector	Industry	Weight	BL input
39	REGN	Health Care	Not classified	0.892%	4.2%
40	AZN.L	Health Care	Not classified	0.858%	3.8%
41	CLS.TO	Information Technology	Electronics Manufacturing Services	0.842%	9.3%
42	KER.PA	Consumer Discretionary	Not classified	0.839%	5.2%
43	AMGN	Health Care	Not classified	0.830%	4.7%
44	SMIN.L	Industrials	Not classified	0.811%	4.6%
45	IMO.TO	Energy	Not classified	0.796%	4.2%
46	NFLX	Communication Services	Not classified	0.789%	2.7%
47	RHHBY	Health Care	Not classified	0.754%	3.6%
48	PDD	Consumer Discretionary	Not classified	0.744%	5.4%
49	WBD	Communication Services	Not classified	0.734%	2.7%
50	TD.TO	Financials	Banking	0.727%	3.8%
51	BAC	Financials	Not classified	0.712%	4.1%
52	PFE	Health Care	Not classified	0.711%	4.4%
53	RR.L	Industrials	Not classified	0.689%	5.9%
54	SAN.PA	Health Care	Not classified	0.669%	3.4%
55	TSLA	Consumer Discretionary	Not classified	0.657%	5.5%
56	TTE.PA	Energy	Not classified	0.647%	2.3%
57	ENEL.MI	Utilities	Not classified	0.646%	3.4%
58	ISP.MI	Financials	Not classified	0.645%	4.4%
59	MC.PA	Consumer Discretionary	Not classified	0.644%	3.9%
60	PEP	Consumer Staples	Not classified	0.640%	2.2%
61	MUFG	Financials	Not classified	0.635%	5.6%
62	C	Financials	Not classified	0.632%	5.1%
63	ABX.TO	Materials	Not classified	0.604%	8.4%
64	CTSH	Information Technology	Not classified	0.591%	3.1%
65	HO.PA	Industrials	Not classified	0.575%	3.6%
66	PCAR	Industrials	Not classified	0.563%	5.2%
67	MCD	Consumer Discretionary	Not classified	0.562%	2.4%
68	FTS.TO	Utilities	Regulated Utilities	0.523%	1.3%
69	SIE.DE	Consumer Discretionary	Not classified	0.522%	5.0%
70	ADI	Information Technology	Not classified	0.511%	6.1%
71	CPG.L	Consumer Staples	Not classified	0.507%	3.8%
72	AD.AS	Consumer Discretionary	Not classified	0.507%	1.9%
73	META	Communication Services	Not classified	0.506%	3.7%
74	ENI.MI	Energy	Not classified	0.461%	2.5%
75	NA.TO	Financials	Banking	0.455%	3.1%
76	CMCSA	Communication Services	Not classified	0.443%	3.0%

Maximum Sharpe allocation schedule 3/3

All positions with absolute portfolio weight of at least 0.005%

Rank	Ticker	Sector	Industry	Weight	BL input
77	ALV.DE	Financials	Not classified	0.411%	3.1%
78	FANG	Energy	Not classified	0.408%	1.9%
79	CSX	Industrials	Not classified	0.398%	4.1%
80	CVS	Health Care	Not classified	0.397%	2.4%
81	SHEL.L	Energy	Not classified	0.389%	2.3%
82	MRK	Health Care	Not classified	0.386%	3.8%
83	MNST	Consumer Staples	Not classified	0.378%	2.1%
84	SN.L	Health Care	Not classified	0.372%	1.8%
85	ABBV	Health Care	Not classified	0.325%	2.5%
86	INTC	Information Technology	Not classified	0.298%	6.8%
87	ZURN.SW	Financials	Not classified	0.284%	2.7%
88	DG.PA	Industrials	Not classified	0.268%	3.6%
89	BATS.L	Consumer Staples	Not classified	0.259%	2.3%
90	XOM	Energy	Not classified	0.257%	1.6%
91	SO	Utilities	Not classified	0.215%	1.6%
92	BHP	Materials	Not classified	0.211%	6.6%
93	LLOY.L	Financials	Not classified	0.204%	4.4%
94	NG.L	Utilities	Not classified	0.185%	2.6%
95	AI.PA	Industrials	Not classified	0.175%	2.3%
96	BRK-B	Financials	Not classified	0.169%	1.5%
97	UBER	Information Technology	Not classified	0.158%	3.0%
98	FNV.TO	Materials	Not classified	0.118%	6.2%
99	TM	Consumer Discretionary	Not classified	0.110%	4.4%
100	MDT	Health Care	Not classified	0.105%	2.7%
101	USB	Financials	Not classified	0.068%	3.7%
102	IBE.MC	Utilities	Not classified	0.057%	1.9%
103	WFC	Financials	Not classified	0.042%	3.7%
104	LMT	Industrials	Not classified	0.036%	2.5%
105	ABI.BR	Consumer Staples	Not classified	0.033%	2.2%
106	EMA.TO	Utilities	Not classified	0.028%	0.5%
107	AAL.L	Materials	Not classified	0.025%	6.5%
108	AMAT	Information Technology	Not classified	0.016%	7.6%

APPENDIX

Minimum Volatility allocation schedule 1/7

All positions with absolute portfolio weight of at least 0.005%

Rank	Ticker	Sector	Industry	Weight	BL input
1	EMA.TO	Utilities	Not classified	1.174%	0.5%
2	EA	Communication Services	Not classified	1.109%	0.4%
3	BCE.TO	Communication Services	Not classified	1.025%	0.0%
4	FTS.TO	Utilities	Regulated Utilities	1.015%	1.3%
5	GWO.TO	Financials	Not classified	1.013%	2.3%
6	TOU.TO	Energy	Not classified	0.921%	0.3%
7	CVX	Energy	Not classified	0.917%	0.2%
8	MRU.TO	Consumer Discretionary	Not classified	0.916%	-0.1%
9	FANG	Energy	Not classified	0.896%	1.9%
10	XOM	Energy	Not classified	0.886%	1.6%
11	RCI-B.TO	Communication Services	Not classified	0.876%	0.2%
12	WN.TO	Consumer Discretionary	Not classified	0.871%	0.4%
13	SU.TO	Energy	Integrated Energy	0.863%	5.4%
14	BP.L	Energy	Not classified	0.833%	3.1%
15	IBE.MC	Utilities	Not classified	0.829%	1.9%
16	IFC.TO	Financials	Not classified	0.823%	-0.0%
17	TTE.PA	Energy	Not classified	0.820%	2.3%
18	EXC	Utilities	Not classified	0.809%	0.3%
19	TMUS	Communication Services	Not classified	0.809%	-3.1%
20	T	Communication Services	Not classified	0.808%	-1.3%
21	DUK	Utilities	Not classified	0.807%	0.9%
22	CNQ.TO	Energy	Integrated Energy	0.802%	5.9%
23	KO	Consumer Staples	Not classified	0.795%	0.6%
24	BRK-B	Financials	Not classified	0.795%	1.5%
25	PPL.TO	Energy	Not classified	0.782%	-1.7%
26	WBD	Communication Services	Not classified	0.779%	2.7%
27	POW.TO	Financials	Not classified	0.773%	3.6%
28	ULVR.L	Consumer Staples	Not classified	0.771%	0.5%
29	ENB.TO	Energy	Energy Infrastructure	0.763%	-2.6%
30	SHEL.L	Energy	Not classified	0.760%	2.3%
31	NESN.SW	Consumer Staples	Not classified	0.749%	0.7%
32	VZ	Communication Services	Not classified	0.746%	-0.6%
33	T.TO	Communication Services	Not classified	0.734%	0.7%
34	ENI.MI	Energy	Not classified	0.725%	2.5%
35	SO	Utilities	Not classified	0.725%	1.6%
36	TRP.TO	Energy	Energy Infrastructure	0.704%	-2.9%
37	MNST	Consumer Staples	Not classified	0.700%	2.1%
38	MO	Consumer Staples	Not classified	0.700%	-0.3%

APPENDIX

Minimum Volatility allocation schedule 2/7

All positions with absolute portfolio weight of at least 0.005%

Rank	Ticker	Sector	Industry	Weight	BL input
39	COST	Consumer Discretionary	Not classified	0.697%	1.2%
40	JNJ	Health Care	Not classified	0.695%	-0.3%
41	IMB.L	Consumer Staples	Not classified	0.689%	1.3%
42	FFH.TO	Financials	Not classified	0.687%	4.6%
43	BNS.TO	Financials	Banking	0.682%	5.4%
44	COP	Energy	Not classified	0.679%	1.8%
45	AD.AS	Consumer Discretionary	Not classified	0.679%	1.9%
46	PEP	Consumer Staples	Not classified	0.679%	2.2%
47	AEP	Utilities	Not classified	0.667%	1.8%
48	AI.PA	Industrials	Not classified	0.659%	2.3%
49	TTWO	Communication Services	Not classified	0.652%	0.8%
50	SN.L	Health Care	Not classified	0.644%	1.8%
51	CVE.TO	Energy	Not classified	0.635%	6.1%
52	CVS	Health Care	Not classified	0.633%	2.4%
53	UBSG.SW	Financials	Not classified	0.632%	3.9%
54	NVDA	Information Technology	Not classified	0.628%	-1.2%
55	PG	Consumer Staples	Not classified	0.627%	0.2%
56	LMT	Industrials	Not classified	0.627%	2.5%
57	HO.PA	Industrials	Not classified	0.618%	3.6%
58	NTR.TO	Materials	Fertilizers & Agricultural Inputs	0.600%	5.6%
59	AVGO	Information Technology	Not classified	0.585%	0.2%
60	GSK.L	Health Care	Not classified	0.581%	4.2%
61	ZURN.SW	Financials	Not classified	0.578%	2.7%
62	MUV2.DE	Financials	Not classified	0.575%	1.6%
63	WMT	Consumer Discretionary	Not classified	0.572%	1.5%
64	MSFT	Information Technology	Not classified	0.569%	-1.4%
65	RY.TO	Financials	Banking	0.568%	4.1%
66	SLF.TO	Financials	Not classified	0.564%	1.2%
67	XEL	Utilities	Not classified	0.558%	1.8%
68	DSY.PA	Industrials	Not classified	0.556%	-2.8%
69	BIP-UN.TO	Industrials	Not classified	0.556%	1.1%
70	LIN	Industrials	Not classified	0.555%	-0.2%
71	BNP.PA	Financials	Not classified	0.552%	5.1%
72	CS.PA	Financials	Not classified	0.535%	2.2%
73	ENGI.PA	Utilities	Not classified	0.533%	5.2%
74	DGE.L	Consumer Staples	Not classified	0.533%	-0.8%
75	ABBV	Health Care	Not classified	0.530%	2.5%
76	NA.TO	Financials	Banking	0.529%	3.1%

APPENDIX

Minimum Volatility allocation schedule 3/7

All positions with absolute portfolio weight of at least 0.005%

Rank	Ticker	Sector	Industry	Weight	BL input
77	DDOG	Information Technology	Not classified	0.528%	-4.4%
78	NG.L	Utilities	Not classified	0.526%	2.6%
79	NFLX	Communication Services	Not classified	0.526%	2.7%
80	ALV.DE	Financials	Not classified	0.521%	3.1%
81	TEAM	Information Technology	Not classified	0.518%	-8.1%
82	KHC	Consumer Staples	Not classified	0.516%	-0.8%
83	IMO.TO	Energy	Not classified	0.515%	4.2%
84	NEE	Utilities	Not classified	0.511%	0.7%
85	VRSK	Financials	Not classified	0.506%	-3.3%
86	ITX.MC	Consumer Discretionary	Not classified	0.504%	2.3%
87	ENEL.MI	Utilities	Not classified	0.500%	3.4%
88	KDP	Consumer Staples	Not classified	0.497%	0.8%
89	REGN	Health Care	Not classified	0.489%	4.2%
90	TD.TO	Financials	Banking	0.486%	3.8%
91	MCD	Consumer Discretionary	Not classified	0.485%	2.4%
92	WDAY	Information Technology	Not classified	0.484%	-4.4%
93	REL.L	Industrials	Not classified	0.479%	-0.2%
94	BATS.L	Consumer Staples	Not classified	0.475%	2.3%
95	BIIB	Health Care	Not classified	0.474%	3.9%
96	AZN.L	Health Care	Not classified	0.465%	3.8%
97	DTE.DE	Communication Services	Not classified	0.464%	-1.2%
98	AAPL	Information Technology	Not classified	0.464%	2.5%
99	QSR.TO	Consumer Discretionary	Not classified	0.462%	3.5%
100	INTU	Information Technology	Not classified	0.451%	-4.8%
101	WCN.TO	Industrials	Waste Management & Environmental Serv	0.451%	-2.1%
102	VOD.L	Communication Services	Not classified	0.441%	1.6%
103	V	Financials	Not classified	0.438%	-0.8%
104	SCHW	Financials	Not classified	0.437%	0.9%
105	ADP	Information Technology	Not classified	0.434%	-2.2%
106	MDLZ	Consumer Staples	Not classified	0.432%	-0.2%
107	MA	Financials	Not classified	0.431%	-0.8%
108	ORCL	Information Technology	Not classified	0.424%	0.1%
109	CL	Consumer Staples	Not classified	0.419%	-0.8%
110	PAYX	Information Technology	Not classified	0.417%	-1.4%
111	OR.PA	Consumer Staples	Not classified	0.416%	1.8%
112	AIG	Financials	Not classified	0.415%	1.5%
113	SAN.PA	Health Care	Not classified	0.413%	3.4%
114	NOVN.SW	Health Care	Not classified	0.406%	4.6%

APPENDIX

Minimum Volatility allocation schedule 4/7

All positions with absolute portfolio weight of at least 0.005%

Rank	Ticker	Sector	Industry	Weight	BL input
115	DOL.TO	Consumer Discretionary	Not classified	0.406%	0.2%
116	EL.PA	Health Care	Not classified	0.393%	1.9%
117	CM.TO	Financials	Banking	0.391%	6.6%
118	ASML.AS	Information Technology	Not classified	0.389%	2.1%
119	META	Communication Services	Not classified	0.389%	3.7%
120	ABI.BR	Consumer Staples	Not classified	0.385%	2.2%
121	MU	Information Technology	Not classified	0.383%	7.1%
122	RIO.L	Materials	Not classified	0.382%	5.8%
123	MDT	Health Care	Not classified	0.377%	2.7%
124	RTX	Industrials	Not classified	0.375%	2.0%
125	MFC.TO	Financials	Insurance	0.373%	7.8%
126	ABT	Health Care	Not classified	0.373%	4.4%
127	GEV	Industrials	Not classified	0.369%	-0.8%
128	CRM	Information Technology	Not classified	0.368%	-2.3%
129	DE	Industrials	Not classified	0.356%	5.1%
130	ROP	Information Technology	Not classified	0.355%	-0.5%
131	UBER	Information Technology	Not classified	0.353%	3.0%
132	GIB-A.TO	Information Technology	Not classified	0.351%	2.8%
133	FTNT	Information Technology	Not classified	0.347%	-1.8%
134	GILD	Health Care	Not classified	0.339%	2.1%
135	GLEN.L	Materials	Not classified	0.336%	4.2%
136	BMO.TO	Financials	Banking	0.336%	1.5%
137	SMIN.L	Industrials	Not classified	0.331%	4.6%
138	PHIA.AS	Health Care	Not classified	0.327%	4.4%
139	ROST	Consumer Discretionary	Not classified	0.325%	6.6%
140	SPG	Consumer Discretionary	Not classified	0.324%	2.4%
141	PANW	Information Technology	Not classified	0.323%	-4.0%
142	TRI.TO	Information Technology	Not classified	0.322%	-5.9%
143	ALNY	Health Care	Not classified	0.321%	5.6%
144	PM	Consumer Staples	Not classified	0.320%	2.0%
145	HSBA.L	Financials	Not classified	0.318%	2.8%
146	DG.PA	Industrials	Not classified	0.316%	3.6%
147	BNY	Financials	Not classified	0.315%	3.1%
148	BKR	Energy	Not classified	0.305%	2.0%
149	CPRT	Consumer Discretionary	Not classified	0.305%	0.3%
150	CSCO	Information Technology	Not classified	0.304%	3.2%
151	CMCSA	Communication Services	Not classified	0.303%	3.0%
152	AMT	Communication Services	Not classified	0.297%	0.4%

APPENDIX

Minimum Volatility allocation schedule 5/7

All positions with absolute portfolio weight of at least 0.005%

Rank	Ticker	Sector	Industry	Weight	BL input
153	CPG.L	Consumer Staples	Not classified	0.295%	3.8%
154	ORLY	Consumer Discretionary	Not classified	0.291%	0.3%
155	SU.PA	Energy	Not classified	0.290%	4.4%
156	BAC	Financials	Not classified	0.290%	4.1%
157	ISP.MI	Financials	Not classified	0.289%	4.4%
158	AIR.PA	Industrials	Not classified	0.285%	1.8%
159	CNR.TO	Industrials	Railroads	0.279%	4.3%
160	ADBE	Information Technology	Not classified	0.279%	-2.1%
161	UNP	Industrials	Not classified	0.279%	1.9%
162	RHHBY	Health Care	Not classified	0.275%	3.6%
163	FAST	Industrials	Not classified	0.273%	1.7%
164	GOOG	Communication Services	Not classified	0.270%	2.7%
165	VRTX	Health Care	Not classified	0.268%	1.6%
166	AMZN	Consumer Discretionary	Not classified	0.257%	2.6%
167	GOOGL	Communication Services	Not classified	0.250%	2.9%
168	AXP	Financials	Not classified	0.244%	2.3%
169	CTAS	Industrials	Not classified	0.235%	0.7%
170	GD	Industrials	Not classified	0.233%	1.8%
171	INSM	Health Care	Not classified	0.226%	2.1%
172	STX	Information Technology	Not classified	0.223%	1.7%
173	RHM.DE	Industrials	Not classified	0.220%	7.9%
174	CRWD	Information Technology	Not classified	0.213%	-2.4%
175	SAF.PA	Industrials	Not classified	0.212%	3.4%
176	EBAY	Consumer Discretionary	Not classified	0.212%	1.2%
177	TMO	Health Care	Not classified	0.208%	2.4%
178	ZS	Information Technology	Not classified	0.204%	-5.1%
179	BKNG	Consumer Discretionary	Not classified	0.204%	0.9%
180	TSLA	Consumer Discretionary	Not classified	0.201%	5.5%
181	PLTR	Information Technology	Not classified	0.200%	1.2%
182	LSEG.L	Financials	Not classified	0.200%	5.5%
183	TSM	Information Technology	Not classified	0.194%	-0.4%
184	UPS	Industrials	Not classified	0.194%	2.3%
185	WFC	Financials	Not classified	0.193%	3.7%
186	APP	Communication Services	Not classified	0.193%	4.2%
187	AMAT	Information Technology	Not classified	0.189%	7.6%
188	SAP.DE	Information Technology	Not classified	0.186%	-0.8%
189	SNPS	Information Technology	Not classified	0.185%	-0.5%
190	HON	Industrials	Not classified	0.185%	1.5%

APPENDIX

Minimum Volatility allocation schedule 6/7

All positions with absolute portfolio weight of at least 0.005%

Rank	Ticker	Sector	Industry	Weight	BL input
191	CSU.TO	Information Technology	Vertical Market Software	0.180%	-0.7%
192	JPM	Financials	Not classified	0.179%	3.2%
193	TGT	Consumer Discretionary	Not classified	0.177%	0.9%
194	NOW	Information Technology	Not classified	0.173%	-5.8%
195	USB	Financials	Not classified	0.171%	3.7%
196	PCAR	Industrials	Not classified	0.168%	5.2%
197	MC.PA	Consumer Discretionary	Not classified	0.167%	3.9%
198	WSP.TO	Industrials	Not classified	0.161%	2.5%
199	MRK	Health Care	Not classified	0.158%	3.8%
200	LLOY.L	Financials	Not classified	0.156%	4.4%
201	KLAC	Information Technology	Not classified	0.153%	8.8%
202	CSX	Industrials	Not classified	0.151%	4.1%
203	DHR	Health Care	Not classified	0.150%	4.4%
204	GM	Consumer Discretionary	Not classified	0.143%	3.0%
205	AMD	Information Technology	Not classified	0.140%	5.9%
206	DIS	Communication Services	Not classified	0.139%	2.5%
207	STLAM.MI	Consumer Discretionary	Not classified	0.131%	2.6%
208	TXN	Information Technology	Not classified	0.130%	4.5%
209	CP.TO	Industrials	Railroads	0.126%	2.2%
210	RACE.MI	Consumer Discretionary	Not classified	0.123%	-1.0%
211	MS	Financials	Not classified	0.120%	4.0%
212	SIE.DE	Consumer Discretionary	Not classified	0.118%	5.0%
213	PFE	Health Care	Not classified	0.115%	4.4%
214	UNH	Health Care	Not classified	0.113%	2.9%
215	BMY	Health Care	Not classified	0.109%	5.6%
216	BA	Industrials	Not classified	0.106%	3.8%
217	IDXX	Health Care	Not classified	0.105%	1.2%
218	DXCM	Health Care	Not classified	0.102%	0.8%
219	OTEX.TO	Information Technology	Not classified	0.100%	2.2%
220	ISRG	Health Care	Not classified	0.099%	2.6%
221	CTSH	Information Technology	Not classified	0.096%	3.1%
222	INTC	Information Technology	Not classified	0.089%	6.8%
223	MAR	Consumer Discretionary	Not classified	0.081%	3.0%
224	RMS.PA	Consumer Discretionary	Not classified	0.076%	2.9%
225	LOW	Consumer Discretionary	Not classified	0.070%	2.3%
226	C	Financials	Not classified	0.067%	5.1%
227	GS	Financials	Not classified	0.066%	2.8%
228	SBUX	Consumer Discretionary	Not classified	0.048%	2.3%

APPENDIX

Minimum Volatility allocation schedule 7/7

All positions with absolute portfolio weight of at least 0.005%

Rank	Ticker	Sector	Industry	Weight	BL input
229	GE	Industrials	Not classified	0.043%	4.1%
230	CDNS	Consumer Discretionary	Not classified	0.043%	1.2%
231	MRVL	Information Technology	Not classified	0.042%	7.8%
232	AKAM	Information Technology	Not classified	0.040%	-1.2%
233	LLY	Health Care	Not classified	0.039%	3.3%
234	DOW	Materials	Not classified	0.039%	8.6%
235	CAT	Industrials	Not classified	0.033%	4.4%
236	NKE	Consumer Discretionary	Not classified	0.032%	1.9%
237	ABNB	Consumer Discretionary	Not classified	0.024%	1.1%
238	AMGN	Health Care	Not classified	0.011%	4.7%
239	MMM	Industrials	Not classified	0.007%	2.9%